Run report: order-332 class at q = 7 (addendum), run 20261013T0347Z

Descriptive record of one run. Interpretation is in the study `README.md` / `REPORT.pdf`.

What ran

- Command: `rustc --edition 2021 -O genus3_build.rs && ./genus3_build 7 3 2 332`.
- Purpose: build the order-332 (= 4·83) isogeny class at q = 7. The frozen first-in-index selection of run 20261013T0320Z put all five of its q = 7 cases in the order-356 class; a target-order argument (the 4th) was added to `genus3_build.rs` in this commit to reach the other 4·prime class near $q^3 + 1 = 344$.
- Regression: with no target argument, `./genus3_build 7 3 2` reproduces the q = 7 records of run 20261013T0320Z byte for byte (checked before this run).
- Spec: the construction and checks are those of `PROTOCOL.md` (unchanged); the target-order selection is the one addition, defaulting to off.
- Isolation: none; shared host. No timing is a result. Values are exact counts, deterministic.
- Raw records: `results/run-20261013T0347Z.jsonl` (`rec` = case, search).

```
run: 20261013T0347Z
purpose: addendum to run-20261013T0320Z -- the order-332 (4*83) isogeny class at q=7 that the frozen first-
in-index selection skipped (all five q=7 cases there landed in the order-356 class)
command: rustc --edition 2021 -O genus3_build.rs && ./genus3_build 7 3 2 332
commit (base): 7caf29236e26456ae1d81d511234b63e651617e9
source sha256: a10f0d2d08fa4f82aaf9018f0988efa90d5bdfba3b2a3247763c7abe7b461552
rustc: rustc 1.93.1 (01f6ddf75 2026-02-11) (Homebrew)
host: Darwin 25.6.0 arm64, Apple M4 Pro, 14 cpus
note: target-order argument added in this commit; original './genus3_build 7 3 2' reproduces
run-20261013T0320Z byte for byte (regression checked above)
uptime at start: 20:53  up 1 day, 20:32, 8 users, load averages: 130.49 128.58 117.15
args 7 3 2 332: real 0.22
uptime at end: 20:53  up 1 day, 20:32, 8 users, load averages: 130.49 128.58 117.15
regenerated after PR review: counters renamed and completed (points_skipped ->
e0_affine_points_nonsquare_uu1 + e0_ramification_points_y0_zero + 2 infinity), pushforward_on_E0 ->
projected_points_on_E0, E0 Weierstrass model added. Original source sha256
63f7367384ea98643d1acf21f4622450ed5b60ac108bbdcedfc6e6b010371d79; all previously recorded fields reproduced
identically. Timings above are from the original run.
```

Recorded values

Requested 3 split and 2 irreducible of order 332; the (b, c) space was exhausted. Found: **2 irreducible, 0 split**.
`candidates_examined` 65432, `linearly_dependent_skipped` 11232.

u (idx b, c)	roots in K	#E0(K)	t_E	in F_q	smooth	N1,N2,N3	#Jac = #E0	divisor points on quartic / tested
49, 8	0	332 = 4·83	12	yes	yes	8, 50, 308	yes	912 / 912
49, 9	0	332 = 4·83	12	yes	yes	8, 50, 308	yes	912 / 912

Both cases: `N` equals `N_predicted = (q+1, q²+1, q³+1−3t_E)`, `N3` also counted on the K model and equal, every transfer divisor F_q-rational with matching Frobenius (912 of 912 points each). P1–P4 hold.

Observation

Exhausting the parametrization `u = t² + bt + c`, the order-332 class has exactly two irreducible-u representatives meeting the genus-3 conditions and no split-u representative. (The order-356 run was not exhaustive — it stopped at the requested 3 split + 2 irreducible — so this is not a comparison of multiplicities between the two classes.)

Missing data

- As in the parent run: no divisor-class arithmetic, so the transfer is not executed as a homomorphism; E0 points not tested, exactly accounted: 152 tested + 178 affine points with `u u'` zero or non-square (`e0_affine_points_nonsquare_uu1`) + 0 with `y0 = 0` + 2 at infinity = 332 per case; `projected_points_on_E0` only checks the E0 equation, not the pushforward.